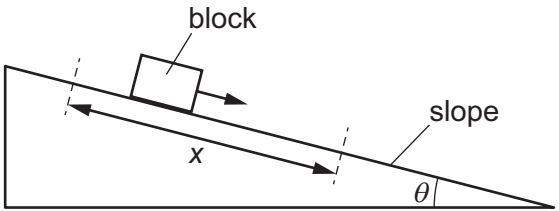
17 A block is released from rest and slides a distance x down a straight slope.

The slope is at an angle θ to the horizontal.



The slope is frictionless and air resistance is negligible.

The acceleration of free fall is g.

Which expression gives the final speed of the block?

A $\sqrt{\frac{gx \sin\theta}{2}}$ **B** $\sqrt{2gx \sin\theta}$ **C** $\sqrt{2gx}$ **D** $2gx \sin\theta$